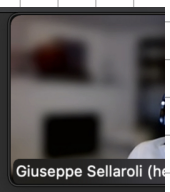
4. Explicit, rigorous definition of $[\Psi]$ tied to Paper 2



In Paper 2 you already have:

- A rank-2 Hermitian bundle $E \to M$.
- A "pure state" section $\Psi \in \Gamma(E)$ with $\|\Psi(x)\| = 1$ in each fiber.
- The density operator field $\rho(x) = |\Psi(x)\rangle\langle\Psi(x)|$.

At each point $x \in M$, the fiber E_x is a 2D complex Hilbert space, isomorphic to $\mathbb{C}^2$.

The set of **pure states** in that fiber is the projective space:
$\mathbb{P}(E_x) \cong \mathbb{CP}^1$.

So you naturally define:

$$[\Psi] : M \to \mathbb{CP}^1, \quad x \mapsto \text{ray of } \Psi(x),$$

i.e. the equivalence class of $\Psi(x)$ modulo nonzero complex scalar.

In a local orthonormal frame (e_1, e_2) of E:
$\Psi(x) = z_1(x)e_1(x) + z_2(x)e_2(x)$,

then
$[\Psi](x) = [z_1(x) : z_2(x)] \in \mathbb{CP}^1$.

This is **exactly** the usual identification of a normalized 2-component spinor with a point on the Bloch sphere.

From there, the Fubini–Study 2-form ω_{FS} on $\mathbb{CP}^1$ is the **unique** U(2)-invariant Kähler form, and physically it's the "distinguishability metric" on pure states (same thing that appears in quantum Fisher information, etc.).

$$\mathcal{H} = \mathbb{C}^2 \qquad P\mathcal{H} = \mathbb{CP}^1 \equiv P(\mathbb{C}^2)$$

$$P \times_{U(1)} \mathbb{C}^2 \to P \times_{U(1)} \mathbb{CP}^1$$

$$\|\Psi(x)\|^2 = 1 \Rightarrow \Psi(x) \neq 0 \quad \forall x \in M$$

$$\Psi(x) = z_1(x)\, e_1(x) + z_2(x)\, e_2(x)$$

$z_i : U \to \mathbb{C}$

e_1, e_2 local orthonormal frame

$e_i : U \subseteq M \to P \times_{U(1)} \mathbb{C}^2$ sections

$\langle e_i|_x, e_j|_x \rangle = \delta_{ij}$

$$\|\Psi\| = 1 \Rightarrow |z_1(x)|^2 + |z_2(x)|^2 = 1$$

$$[\Psi] : M \to \mathbb{CP}^1$$
$$x \mapsto [\Psi(x)]$$

$$[(z,w)] = \{\lambda(z,w) \mid \lambda \in \mathbb{C} - \{0\}\}$$

<u>immersion</u>: $\boxed{d[\Psi]_x \text{ is injective for each } x}$

$\|\Psi(x)\|^2 = 1 = |z_1|^2 + |z_2|^2$ in local coordinates

$$0 = \partial_\mu(|z_1|^2 + |z_2|^2) = (\partial_\mu \bar{z}_1) z_1 + \bar{z}_1 \partial_\mu z + (\partial_\mu \bar{z}_2) z_2 + \bar{z}_2 \partial_\mu z_2$$

$\mathbb{CP}^1$ local coordinates

$z_2 \neq 0 \qquad [(z_1, z_2)] \mapsto z_1/z_2$ ← local homogeneous coordinates

$$[z_1 : z_2] = [\lambda z_1 : \lambda z_2] = [z_1/z_2 : 1]$$

$$\widehat{[\psi]}(x) = \varphi(x) \qquad \varphi(x) = z_1(x)/z_2(x)$$

$$d[\psi]_x\left(\frac{\partial}{\partial x^\mu}\right) = \frac{\partial \varphi}{\partial x^\mu}\frac{\partial}{\partial z} + \frac{\partial \bar\varphi}{\partial x^\mu}\frac{\partial}{\partial \bar z}$$

$$\partial_\mu \varphi = \frac{z_2 \partial_\mu z_1 - z_1 \partial_\mu z_2}{z_2^2} \qquad \partial_\mu \bar\varphi = \frac{\bar z_2 \partial_\mu \bar z_1 - \bar z_1 \partial_\mu \bar z_2}{\bar z_2^2}$$

$$\partial_\mu \varphi \cdot \partial_\mu \bar\varphi = \frac{|z_2|^2 |\partial_\mu z_1|^2 - \bar z_1 z_2 \partial_\mu z_1 \partial_\mu \bar z_2 - \bar z_2 z_1 \partial_\mu z_2 \partial_\mu \bar z_1 + |z_1|^2 |\partial_\mu z_2|^2}{|z_2|^4}$$

$$|z_1|^2 + |z_2|^2 = 1 \qquad \varphi = \frac{z_1}{z_2} \qquad |\varphi|^2 = \frac{|z_1|^2}{|z_2|^2} = \frac{1 - |z_2|^2}{|z_2|^2}$$

$$\partial_\mu |\varphi|^2 = \bar\varphi \partial_\mu \varphi + \varphi \partial_\mu \bar\varphi$$

$$\| $$

$$\frac{|z_2|^2 \partial_\mu |z_2|^2 - (1 - |z_2|^2)\partial_\mu |z_2|^2}{|z_2|^4} = \frac{2|z_2|^2 \partial_\mu |z_2|^2 - 1}{|z_2|^4}$$

<u>ex:</u> $\quad z_1(x) = 0 \qquad z_2(x) = 1 \qquad \|\psi\| = 1$

$[\psi](x)$ has coordinates $\dfrac{z_1(x)}{z_2(x)} = 0$ ← constant

so $d[\psi]_x = 0$

$$\theta = d(\arg f) \qquad f = z_1 \cdot z_2$$

$$\arg f = \arg(z_1) + \arg(z_2) \quad (\text{mod } 2\pi)$$

$$d(\arg f) = d(\arg z_1) + d(\arg z_2)$$

$z_2 = \frac{1}{\sqrt{2}}$ $\quad z_1(x) = \frac{e^{i\alpha(x)}}{\sqrt{2}}$ $\quad \alpha : M \to \mathbb{R}$

$\theta = d\alpha$ never zero $\leftarrow$ exact form

$\theta(T) = 1$ $\quad \Omega \wedge d(i_X \Omega) = 0$ if $X \in \ker(\theta)$

- give θ global nowhere vanishing 1-form
- select any T s.t. $\theta(T) = 1$
- $S = \ker \theta$
- $TM = \mathrm{span}(T) \oplus S = \mathrm{span}(T + X) \oplus S$ for any $X \in S$

$\mathbb{R}^2 = \mathrm{span}(1,0) \oplus \mathrm{span}(0,1)$
$= \mathrm{span}(1,1) \oplus \mathrm{span}(0,1)$

Connection between θ and $\psi \in \Gamma(P \times_{U(1)} \mathbb{C}^2)$ that has some properties

- build $[\psi] : M \to \mathbb{CP}^1$
- build $\Omega = [\psi]^* \omega_{FS}$
- build volume form on M $\quad \underline{\nu = \Omega \wedge d(i_T \Omega)}$

if $X \in \ker \theta$ $\quad \theta(T + X) = \theta(T)$

$\Omega \wedge \Omega = 0$
$= [\psi^*](\omega_{FS} \wedge \omega_{FS})$
$= 0$

$d\omega_{FS} = 0$

$\boxed{d\Omega = 0}$

$$\underbrace{\Omega \wedge d(i_X \Omega)} = d(\underbrace{\Omega \wedge i_X \Omega})$$

$\hookrightarrow$ is it zero?

$d i_X + i_X d = \mathcal{L}_X$

$$\Omega \wedge (\mathcal{L}_X \Omega - \cancel{i_X d\Omega}^{\,0})$$

$\alpha \wedge \beta = (-1)^{|\alpha|} \beta \wedge \alpha$

$$\Omega \wedge \mathcal{L}_X \Omega = ?$$

$$0 = \mathcal{L}_X(\Omega \wedge \Omega) = \mathcal{L}_X \Omega \wedge \Omega + \Omega \wedge \mathcal{L}_X \Omega$$
$$= 2 \mathcal{L}_X \Omega \wedge \Omega$$

$$d\, i_X + i_X\, d = \mathcal{L}_X$$

$$i_X(\alpha\wedge\beta) = i_X\alpha\wedge\beta + (-1)^{|\alpha|}\alpha\wedge i_X\beta$$

$$d(\alpha\wedge\beta) = d\alpha\wedge\beta + (-1)^{|\alpha|}\alpha\wedge d\beta$$

$$d\, i_X(\alpha\wedge\beta) = d\left(i_X\alpha\wedge\beta + (-1)^{|\alpha|}\alpha\wedge i_X\beta\right)$$

$$= d\, i_X\alpha\wedge\beta - (-1)^{|\alpha|} i_X\alpha\wedge d\beta + (-1)^{\alpha} d\alpha\wedge i_X\beta + (-1)^{\alpha}\alpha\wedge d\, i_X\beta$$

$$i_X d(\alpha\wedge\beta) = i_X d\alpha\wedge\beta - (-1)^{|\alpha|} d\alpha\wedge i_X\beta + (-1)^{\alpha} i_X\alpha\wedge d\beta + (-1)^{\alpha}\alpha\wedge i_X d\beta$$

$$\mathcal{L}_X(\alpha\wedge\beta) = \mathcal{L}_X\alpha\wedge\beta + (-1)^{\alpha}\alpha\wedge\mathcal{L}_X\beta$$

$$\nu = \Omega\wedge d(\iota_T\Omega) \qquad \Omega = [\Psi]^*\omega_{FS}$$

$$d(\alpha\wedge\beta) = d\alpha\wedge\beta + (-1)^{|\alpha|}\alpha\wedge d\beta \quad \Rightarrow \quad \boxed{\begin{aligned} d\Omega &= [\Psi^*]\, d\omega_{FS} = 0 \\ \Omega\wedge\Omega &= [\Psi^*](\omega_{FS}\wedge\omega_{FS}) = 0 \end{aligned}}$$

$$d(\underbrace{\Omega}_{\deg 2}\wedge \underbrace{\iota_T\Omega}_{\deg 1}) = \underbrace{\overbrace{d\Omega}^{=0}\wedge\iota_T\Omega}_{=0} + \Omega\wedge d(\iota_T\Omega)$$

$$\alpha\wedge\beta = (-1)^{|\alpha||\beta|}\beta\wedge\alpha$$

so $\nu = d(\Omega\wedge\iota_T\Omega)$

$$\Omega\wedge\Omega = 0 \quad \Rightarrow \quad 0 = \iota_T(\Omega\wedge\Omega) = \iota_T\Omega\wedge\Omega + \Omega\wedge\iota_T\Omega$$

$$= 2\,\Omega\wedge\iota_T\Omega$$

$$\boxed{\nu = 0}$$

$$\Omega = A\, dt\wedge dx + B\, dt\wedge dy + C\, dt\wedge dz + D\, dx\wedge dy + E\, dx\wedge dz + F\, dy\wedge dz$$

$$T = \partial_t$$

$$\iota_T\Omega = A\, dx + B\, dy + C\, dz$$

$$d\iota_T\Omega = \partial_t A\, dt\wedge dx + \partial_y A\, dy\wedge dx + \partial_z A\, dz\wedge dx$$
$$+\partial_t B\, dt\wedge dy + \partial_x B\, dx\wedge dy + \partial_z B\, dz\wedge dy$$
$$+\partial_t C\, dt\wedge dz + \partial_x C\, dx\wedge dz + \partial_y C\, dy\wedge dz$$

$$\Omega\wedge\Omega = 0 \quad \rightarrow \quad AF - BE + CD$$

$$d\Omega = \underline{\partial A_y}^{+}\, dy\wedge dt\wedge dx + \underline{\partial A_z}^{+}\, dz\wedge dt\wedge dx$$
$$\underline{\partial B_x}^{-}\, dx\wedge dt\wedge dy + \underline{\partial B_z}^{+}\, dz\wedge dt\wedge dy$$
$$\underline{\partial C_x}^{-}\, dx\wedge dt\wedge dz + \underline{\partial C_y}^{-}\, dy\wedge dt\wedge dz$$
$$\underline{\partial D_t}^{+}\, dt\wedge dx\wedge dy + \underline{\partial D_z}^{+}\, dz\wedge dx\wedge dy$$
$$\underline{\partial E_t}^{+}\, dt\wedge dx\wedge dz + \underline{\partial E_y}^{-}\, dy\wedge dx\wedge dz$$
$$\underline{\partial F_t}^{+}\, dt\wedge dy\wedge dz + \underline{\partial F_x}^{+}\, dx\wedge dy\wedge dz$$

$$\partial_y A - \partial B_x + \partial D_t = 0 \qquad\qquad AF - BE + CD = 0$$
$$\partial_z A - \partial C_x + \partial E_t = 0$$
$$\partial_z B - \partial C_y + \partial F_t = 0$$
$$\partial D_z - \partial E_y + \partial F_x = 0$$

$$d\iota_T\Omega = dt\wedge dx\ \partial_t A + dt\wedge dy\ \partial_t B + dt\wedge dz\ \partial_t C$$
$$+ dx\wedge dy\ \partial D_t + dy\wedge dz\ \partial F_t + dx\wedge dz\ \partial_t E$$

$$\Omega = A\, dt\wedge dx + B\, dt\wedge dy + C\, dt\wedge dz + D\, dx\wedge dy + E\, dx\wedge dz + F\, dy\wedge dz$$

$$\Omega \wedge d\iota_T \Omega = A\,\partial_t F\, dt\wedge dx\wedge dy\wedge dz - B\,\partial_t E\, dt\wedge dx\wedge dy\wedge dz$$

$$C\,\partial_t D\, dt\wedge dx\wedge dy\wedge dz + D\,\partial_t C\, dt\wedge dx\wedge dy\wedge dz$$

$$-E\,\partial_t B\, dt\wedge dx\wedge dy\wedge dz + F\,\partial_t A\, dt\wedge dx\wedge dy\wedge dz$$

$$\nu = dt\wedge dx\wedge dy\wedge dz\left(\partial_t(AF) + \partial_t(DC) - \partial_t(BE)\right)$$

$$= dt\wedge dx\wedge dy\wedge dz\;\partial_t\big(\underbrace{AF + CD - BE}_{\Omega\wedge\Omega = 0}\big)$$

$\mathbb{C}^n$ is a complex manifold

→ transition functions are holomorphic

$\mathbb{CP}^{n-1}$ is an $(n-1)$-dim complex manifold

holomorphic coordinates → treat it as a $2n$-dimensional real manifold

$(z, \bar{z})$, $z \in \mathbb{C}^n$ treat $z, \bar{z}$ as independent

vector field $X = A^\mu \frac{\partial}{\partial z^\mu} + B_\mu \frac{\partial}{\partial \bar{z}_\mu}$ $\alpha = a_\mu dz^\mu + b^\mu d\bar{z}_\mu$

$h = d\bar{z}_\mu \otimes dz^\mu$ Hermitian form (inner product at each point)

Aside: Wirtinger derivatives $z = x + iy$

$$\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)$$

$$\frac{\partial}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)$$

f satisfies Cauchy-Riemann if $\frac{\partial f}{\partial \bar{z}} = 0$

ex: $f(z,\bar{z}) = |z|^2$ $\frac{\partial f}{\partial \bar{z}} = z$ not holomorphic

$f(z,\bar{z}) = z^2$

$\frac{\partial f}{\partial \bar{z}} = 0$

$\alpha : V \to \mathbb{C}$ $\qquad$ $\alpha \otimes \beta : V \times W \to \mathbb{C}$

$\beta : W \to \mathbb{C}$ $\qquad$ $(v, w) \mapsto \alpha(v)\,\beta(w)$

$\alpha \otimes \beta$ bilinear

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in our case at each point $x \in \mathbb{C}^n$ $\quad dz^\mu|_x : T_x\mathbb{C}^n \to \mathbb{C}$

$$dz^\mu\left(\frac{\partial}{\partial z^\nu}\right) = \delta^\mu_\nu \qquad dz^\mu\left(\frac{\partial}{\partial \bar{z}^\nu}\right) = 0$$

$$dz^\mu \otimes d\bar{z}_\mu|_x : \underbrace{T_x\mathbb{C}^n}_{\cong \mathbb{C}^n} \times T_x\mathbb{C}^n \to \mathbb{C}$$

Complex inner product

$\langle u, v\rangle = \overline{\langle v, u\rangle}$

$\langle u, u\rangle > 0$ if $u \neq 0$

$\langle u, v\rangle = u^* v$

<u>real coordinates</u> $\quad z^\mu = x^\mu + i y^\mu \qquad dz^\mu = dx^\mu + i\,dy^\mu$

$d\bar{z}^\mu = dx^\mu - i\,dy^\mu$

$$\delta_{\mu\nu}\, dz^\mu \otimes d\bar{z}^\nu = \delta_{\mu\nu}(dx^\mu + i\,dy^\mu) \otimes (dx^\nu - i\,dy^\nu)$$

$$= \delta_{\mu\nu}\underbrace{(dx^\mu \otimes dx^\nu + dy^\mu \otimes dy^\nu)}_{g} - i\,\delta_{\mu\nu}\underbrace{(dx^\mu \otimes dy^\nu - dy^\mu \otimes dx^\nu)}_{\omega}$$

$g \to$ symmetric, Pos. def. / Riemann

$\omega \to$ anti symmetric, non-degenerate, symplectic

$n = 1$ $\quad$ vector field:

$\sigma_x \sigma_p \geq \frac{\hbar}{2}$

$$A = A_x \frac{\partial}{\partial x} + A_y \frac{\partial}{\partial y} \qquad B = B_x \frac{\partial}{\partial x} + A_y \frac{\partial}{\partial y}$$

$\leftrightarrow A_x + i A_y$ in $\mathbb{C}$

$$h(A, B) = ? = (dx \otimes dx + dy \otimes dy - i\,dx \otimes dy + i\,dy \otimes dx)(A, B)$$

$$= A_x B_x + A_y B_y - i A_x B_y + i B_x A_y$$

$$= (A_x + i A_y)\,\overline{(B_x + i B_y)}$$

$$= (A_x + i A_y)(B_x - i B_y)$$

<u>complex conjugation</u> $\qquad J_x : T_x\mathbb{C}^n \to T_x\mathbb{C}^n \qquad J^2 = -\mathrm{id}$

$$J\left(\frac{\partial}{\partial x^\mu}\right) = \frac{\partial}{\partial y^\mu} \qquad J\left(\frac{\partial}{\partial y^\mu}\right) = -\frac{\partial}{\partial x^\mu}$$

$$J^2\left(\frac{\partial}{\partial x^\mu}\right) = -\frac{\partial}{\partial x^\mu} \qquad J^2\left(\frac{\partial}{\partial y^\mu}\right) = -\frac{\partial}{\partial y^\mu}$$

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

$$T_a\mathbb{C}^n = \left\{ A^\mu \frac{\partial}{\partial x^\mu}\Big|_a + B^\mu \frac{\partial}{\partial y^\mu}\Big|_a \;\Big|\; A, B \in \mathbb{R}^n \right\}$$

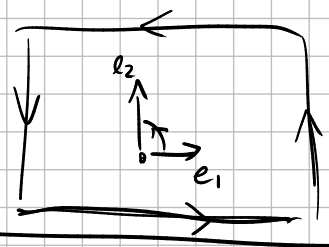
$$\frac{\partial}{\partial z^\mu} = \frac{1}{2}\left(\frac{\partial}{\partial x^\mu} - i\frac{\partial}{\partial y^\mu}\right)$$

$$\frac{\partial}{\partial x^\mu} = \frac{\partial}{\partial z^\mu} + \frac{\partial}{\partial \bar{z}^\mu}$$

$$\frac{\partial}{\partial y^\mu} = i\left(\frac{\partial}{\partial z^\mu} - \frac{\partial}{\partial \bar{z}^\mu}\right)$$

$$A^\mu \frac{\partial}{\partial x^\mu}\Big|_a + B^\mu \frac{\partial}{\partial y^\mu}\Big|_a =$$

$$= \underline{(A^\mu + iB^\mu)} \frac{\partial}{\partial z^\mu} + \underline{(A^\mu - iB^\mu)} \frac{\partial}{\partial \bar{z}^\mu}$$



$$\mathbb{CP}^1 \cong S^2$$

$$\begin{cases} h_a : T_aM \times T_aM \to \mathbb{C} & \text{bilinear} \\ h_a(v, Jw) = -i\, h_a(v, w) \\ h_a(v, w) = \overline{h_a(w, v)} \\ h_a(v, v) > 0 \quad \text{if } v \neq 0 \end{cases}$$

$\to$ Hermitian form on an almost complex manifold (M, J)

$$\bar{h}(v, w) = \overline{h(v, w)}$$

$$g = \frac{1}{2}(h + \bar{h}) : T_aM \times T_aM \to \mathbb{R}$$

$g(u, v) = g(v, u)$

$\begin{cases} \text{symmetric} \\ \text{positive-def} \end{cases} \to$ Riemannian metric

$g(v, v) > 0$ if $v \neq 0$

$$\omega = \frac{1}{2i}(h - \bar{h}) : T_aM \times T_aM \to \mathbb{R}$$

$\begin{cases} \text{anti symmetric} \\ \text{non-degenerate} \end{cases} \to$ pre-symplectic form

$$g(u, v) = \omega(u, Jv) \quad \text{compatibility}$$

$\mathbb{CP}^n$ is a Kahler manifold

$H(\psi, \bar{\psi}) = \langle \psi | \hat{H} | \psi \rangle$

<u>coordinates:</u> $[z_0 : z_1 : \cdots : z_n] \in \mathbb{CP}^n$

if $z_0 \neq 0$ coords: $z_1 = \frac{z_1}{z_0}$, $z_2 = \frac{z_2}{z_0}$, ...

$$h = \frac{(1+\|z\|^2)\,\delta_{\mu\nu} - \bar{z}_\mu z_\nu}{(1+\|z\|^2)^2}\, dz^\mu \otimes d\bar{z}^\nu$$

$$J\left(A^\mu \frac{\partial}{\partial z^\mu} + \bar{A}^\mu \frac{\partial}{\partial \bar{z}^\mu}\right) = iA^\mu \frac{\partial}{\partial z^\mu} - i\bar{A}^\mu \frac{\partial}{\partial \bar{z}^\mu}$$

$$h\left(A^\mu \frac{\partial}{\partial z^\mu} + \bar{A}^\mu \frac{\partial}{\partial \bar{z}^\mu},\ A^\nu \frac{\partial}{\partial z^\nu} + \bar{A}^\nu \frac{\partial}{\partial \bar{z}^\nu}\right)$$

$$= \frac{(1+\|z\|^2)\,\delta_{\mu\nu} - \bar{z}_\mu z_\nu}{(1+\|z\|^2)^2}\left(A^\mu \bar{A}^\nu\right)$$

$$= \frac{1}{\cdots}\left[(1+\|z\|^2)\left(\sum_\mu |A^\mu|^2\right) - |\bar{z}_\mu A^\mu|^2\right]$$

$$\|z\|^2 \|A\|^2 \geq |\bar{z}\cdot A|^2$$

$$\frac{1}{\cdots}\left[\underbrace{\|A\|^2 + \|z\|^2\|A\|^2 - |\bar{z}\cdot A|^2}_{\geq \|A\|^2}\right]$$

$$h\left(A^\mu \frac{\partial}{\partial z^\mu} + \bar{A}^\mu \frac{\partial}{\partial \bar{z}^\mu},\ B^\nu \frac{\partial}{\partial z^\nu} + \bar{B}^\nu \frac{\partial}{\partial \bar{z}^\nu}\right)$$

$$= \frac{(1+\|z\|^2)\,\delta_{\mu\nu} - \bar{z}_\mu z_\nu}{(1+\|z\|^2)^2}\left(A^\mu \bar{B}^\nu\right)$$

$$= \frac{(1+\|z\|^2)\,\delta_{\mu\nu} - \bar{z}_\nu z_\mu}{(1+\|z\|^2)^2}\left(A^\nu \bar{B}^\mu\right)$$

$$= h\left(B^\nu \frac{\partial}{\partial z^\nu} + \bar{B}^\nu \frac{\partial}{\partial \bar{z}^\nu},\ A^\mu \frac{\partial}{\partial z^\mu} + \bar{A}^\mu \frac{\partial}{\partial \bar{z}^\mu}\right)$$

$$h\left(A^\mu \frac{\partial}{\partial z^\mu} - \bar{A}^\mu \frac{\partial}{\partial \bar{z}^\mu} , iB^\nu \frac{\partial}{\partial z^\nu} - i\bar{B}^\nu \frac{\partial}{\partial \bar{z}^\nu} \right)$$

$$= \frac{(1+\|z\|^2)\,\delta_{\mu\nu} - \bar{z}_\mu z_\nu}{(1+\|z\|^2)^2} \left(A^\mu (-i)\, \bar{B}^\nu \right)$$

$$= -ih\left(A^\mu \frac{\partial}{\partial z^\mu} + \bar{A}^\mu \frac{\partial}{\partial \bar{z}^\mu} , B^\nu \frac{\partial}{\partial z^\nu} + \bar{B}^\nu \frac{\partial}{\partial \bar{z}^\nu} \right)$$

$$\omega = \frac{1}{2i}(h - \bar{h}) = \frac{1}{2i} \frac{(1+\|z\|^2)\,\delta_{\mu\nu} - \bar{z}_\mu z_\nu}{(1+\|z\|^2)^2} \overbrace{\left(dz^\mu \otimes d\bar{z}^\nu - d\bar{z}^\nu \otimes dz^\mu \right)}^{dz^\mu \wedge d\bar{z}^\nu}$$

$$= \frac{i}{2} \frac{\bar{z}_\mu z_\nu \, dz^\mu \wedge d\bar{z}^\nu}{(1+\|z\|^2)^2}$$

$$g = \frac{1}{2}(h + \bar{h}) = \frac{1}{2} \frac{\delta_{\mu\nu}}{1+\|z\|^2} \left(dz^\mu \otimes d\bar{z}^\nu + d\bar{z}^\mu \otimes dz^\nu \right)$$

in terms of Z $\qquad Z \to \lambda z$

$$h = \frac{\overset{|\lambda|^2}{\|Z\|^2} \delta_{\mu\nu} - \overset{|\lambda|^2}{\bar{Z}_\mu Z_\nu}}{\underset{\to |\lambda|^4}{\|Z\|^4}} \overset{\lambda}{dZ^\mu} \otimes \overset{\bar{\lambda}}{d\bar{Z}^\nu}$$

$\varphi^* h$ $\qquad \varphi : \mathbb{C}^{n+1} \setminus \{0\} \to \mathbb{P}\mathbb{C}^n$

$Z \mapsto [Z_0 : Z_1 : \cdots : Z_n]$

$$\varphi^* \omega = \frac{i}{2} \frac{\bar{Z}_\mu Z_\nu}{\|Z\|^4} dZ^\mu \wedge d\bar{Z}^\nu$$

$M \to E$ $\qquad E$ bdly $\cong M \times \mathbb{C}^2$

$d\varphi^* \omega = 0$

$\varphi : \mathbb{C}^2 \setminus \{0\} \to \mathbb{C}P^1$

$\psi : M \to E$ section $\qquad \|\psi\| = 1$

$K : M \to \mathbb{C}^2 \setminus \{0\}$

$x \mapsto \psi(x)$

$\Omega = K^* \varphi^* \omega_{FS}$

helpful to avoid local coordinates on $\mathbb{CP}^1$

<u>ex:</u> circle $x^2 + y^2 = 1$ $\to$ $d(x^2+y^2) = 0 = 2x\,dx + 2y\,dy$

ex: metric on circle (pulled back): $dx \otimes dx + dy \otimes dy$

$$= dy \otimes dy \left(1 - \frac{y}{x}\right)$$

$dx = -\frac{y}{x} dy$

$h = g + i\omega$ $\to$ symplectic $\Rightarrow$ dynamics

T^*M has natural symplectic form

$\searrow$ phase space

(q, p)

$\swarrow$ position $\searrow$ momentum

ω symplectic sends functions to vector field

choose special function $H \to X_H$

<u>dynamics:</u> $\dot{\gamma} = X_H$ $\leftarrow$ Hamilton's eq.

$\gamma : I \subseteq \mathbb{R} \to T^*M$

TM : position + velocity

if system is conservative: $H = T + V$ $\to$ $p = mv$

if not true (ex Lorentz force) p does not have to be $= mv$

$\searrow$ charged particle in electromagnetic field

$$F = q(E + B \times v)$$

$$p = mv - qA$$